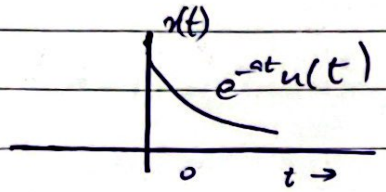
Date

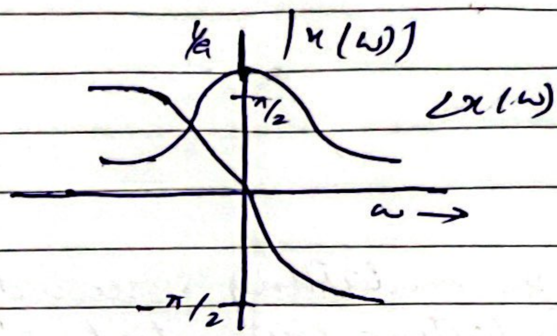
$$\sqrt{a^2+w^2} \quad e^{-j\tan^{-1}\left(\frac{w}{a}\right)}$$

$$x(w) = \frac{1}{\sqrt{a^2+w^2}} e^{-j\tan^{-1}\left(\frac{w}{a}\right)}$$

$$|X(w)| = \frac{1}{\sqrt{a^2+w^2}} \qquad \angle x(w) = -\tan^{-1}\left(\frac{w}{a}\right)$$



$\Longleftrightarrow$



DAY-20 (After MID-2)

Fourier Transform:-

$$x(t) \Longleftrightarrow X(w)$$

$$X(w) = \int_{-\infty}^{\infty} x(t)\, e^{-jwt}\, dt$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} x(w)\, e^{jwt}\, dw$$

Properties of Fourier Transform:-

→ Linearity:-

$$x_1(t) \Longleftrightarrow x_1(w)$$

$$x_2(t) \Longleftrightarrow x_2(w)$$

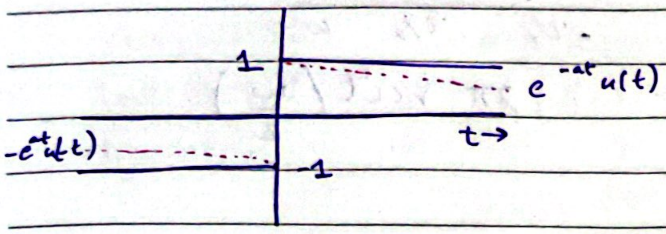
$$a_1 x_1(t) + a_2 x_2(t) \Longleftrightarrow a_1 X_1(w) + a_2 x_2(w)$$

Find the fourier transform for

sgn t (signum t)

Date

$$\operatorname{sgn} t = \begin{cases} 1 & t > 0 \\ -1 & t < 0 \end{cases}$$



$$\operatorname{sgn} t = \lim_{a \to 0} \left[e^{-at}u(t) - e^{at}u(-t) \right]$$

$$\mathcal{F}\{\operatorname{sgn} t\} = \lim_{a \to 0} \left[\mathcal{F}\{e^{-at}u(t)\} - \mathcal{F}\{e^{at}u(-t)\} \right]$$

$$= \lim_{a \to 0} \left[\frac{1}{a+j\omega} - \frac{1}{a-j\omega} \right]$$

$$= \lim_{a \to 0} \left(\frac{-2j\omega}{a^2+\omega^2} \right)$$

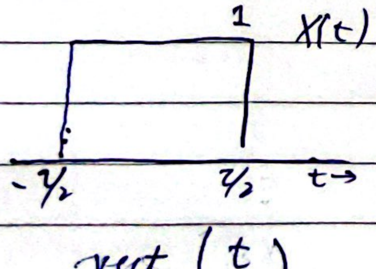
$$= \frac{2}{j\omega}$$

$$\boxed{\operatorname{sgn} t \Leftrightarrow \frac{2}{j\omega}}$$

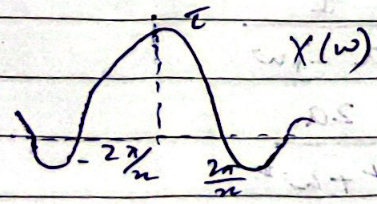
→ **Duality**

$x(t) \Leftrightarrow X(\omega)$

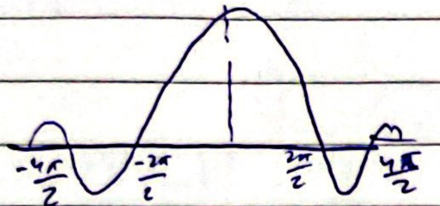
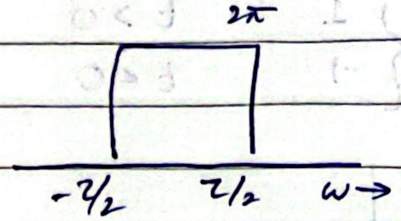
$X(t) \Leftrightarrow 2\pi x(-\omega)$



$\Longleftrightarrow$



$\operatorname{rect}\left(\frac{t}{\tau}\right)$ $\qquad$ $\tau \operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$

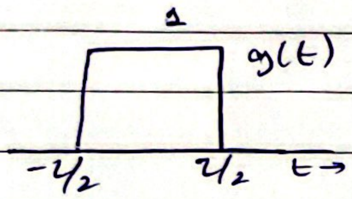
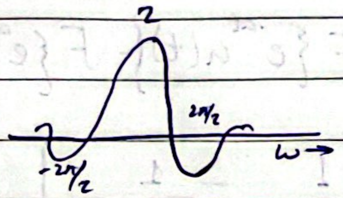
 $\Longleftrightarrow$ 

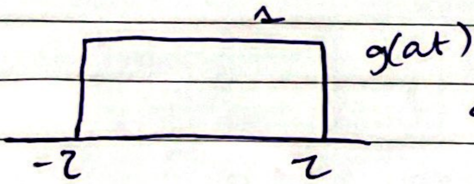
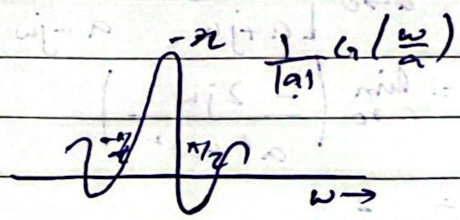
$\tau \, \text{sinc}\left(\frac{\tau t}{2}\right)$ $\qquad$ $2\pi \, \text{rect}\left(\frac{\omega}{\tau}\right)$

→ **Scaling**

$$g(t) \Longleftrightarrow G(\omega)$$

$$g(at) \Longleftrightarrow \frac{1}{|a|} G\left(\frac{\omega}{a}\right)$$

 $\Longleftrightarrow$ 

 $\Longleftrightarrow$ 

DAY-21

Fourier Transforms :-

$x(t)$	$X(\omega)$
$e^{-at}u(t)$	$\frac{1}{a+j\omega}$
$e^{at}u(-t)$	$\frac{1}{a-j\omega}$
$e^{-a\|t\|}$	$\frac{2a}{a^2+\omega^2}$

Date

$\delta(t)$	1
1	$2\pi\,\delta(\omega)$
$e^{j\omega_0 t}$	$2\pi\,\delta(\omega-\omega_0)$
$\cos\omega_0 t$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$
$\sin\omega_0 t$	$j\pi[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)]$
$\operatorname{sgn} t$	$\frac{2}{j\omega}$
$\frac{W}{\pi}\,\operatorname{sinc}(Wt)$	$\operatorname{rect}\left(\frac{\omega}{2W}\right)$

Time Shifting Property :-

$$g(t) \Longleftrightarrow G(\omega)$$
$$g(t-t_0) \Longleftrightarrow G(\omega)\,e^{-j\omega t_0}$$

Frequency Shifting Property :-

$$g(t) \Longleftrightarrow G(\omega)$$
$$g(t)\,e^{j\omega_0 t} \Longleftrightarrow G(\omega-\omega_0)$$

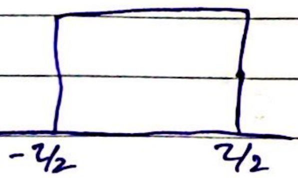
since $e^{j\omega_0 t}$ is not a real function, frequency shifting is achieved by multiplying $g(t)$ with a sinosoid

$$g(t)\cos\omega_0 t = \frac{1}{2}\left[g(t)\,e^{j\omega_0 t} + g(t)\,e^{-j\omega_0 t}\right]$$

$$g(t)\cos\omega_0 t \Longleftrightarrow \frac{1}{2}\left[G(\omega-\omega_0)+G(\omega+\omega_0)\right]$$

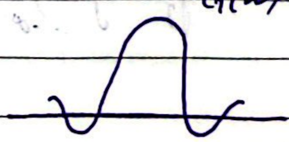
the final result shows that multiplication of signal $g(t)$ by sinusoid of ω_0 shifts spectrum $G(\omega)$ by + and − ω_0

$g(t)$



$-\tau/2$ $\tau/2$

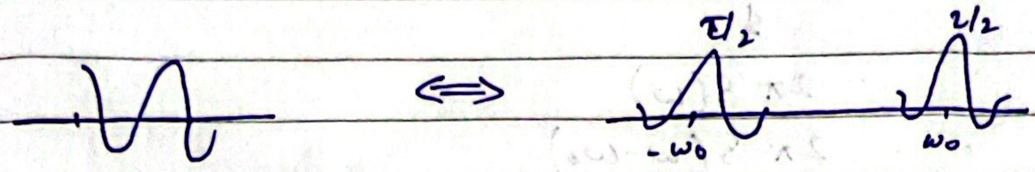
$G(\omega)$

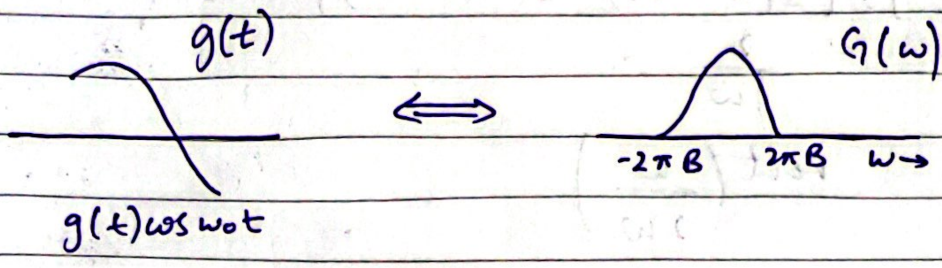


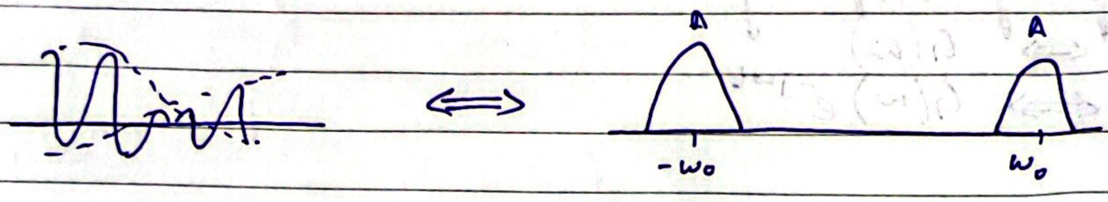
$\tau\,\operatorname{sinc}\left(\frac{\omega\tau}{2}\right)$

$g(t)\cos\omega_0 t$

Date







Convolution :-

$$g(t) * x(t) = \int_{-\infty}^{\infty} g(\tau)\, x(t-\tau)\, d\tau$$

Convolution Property :-

$$g_1(t) * g_2(t) \Longleftrightarrow G_1(\omega)\, G_2(\omega)$$

If $g_1(t) \Longleftrightarrow G_1(\omega)$ and $g_2(t) \Longleftrightarrow G_2(\omega)$

then time convolution

$$g_1(t) * g_2(t) \Longleftrightarrow G_1(\omega)\, G_2(\omega)$$

$$g_1(t)\, g_2(t) \Longleftrightarrow \frac{1}{2\pi} G_1(\omega) * G_2(\omega)$$

Proof of Time convolution :

$$F\left[g_1(t) * g_2(t)\right] = \int_{-\infty}^{\infty} e^{-j\omega t} \left[\int_{-\infty}^{\infty} g_1(t)\, g\right.$$

Date

DAY - 22

Time differentiation and time integration properties :-

$$x(t) \Longleftrightarrow X(\omega)$$

then $\frac{dx}{dt} \Longleftrightarrow j\omega\, X(\omega)$ — time differentiation property

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

differentiate both sides

$$\frac{dx}{dt} = \frac{1}{2\pi}\int_{-\infty}^{\infty} j\omega\, X(\omega)\, e^{j\omega t}\, d\omega$$

$$\frac{dx}{dt} \Longleftrightarrow j\omega\, X(\omega)$$

Repeated application of differentiation yields

$$\frac{d^n x}{dt^n} \Longleftrightarrow (j\omega)^n X(\omega)$$

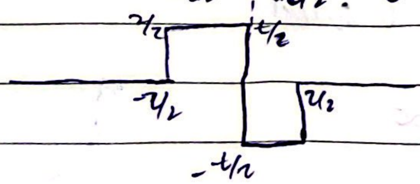
Time integration Property :-

$$\int_{-\infty}^{\infty} x(\tau)\, d\tau \Longleftrightarrow \frac{X(\omega)}{j\omega} + \pi X(0)\,\delta(\omega)$$

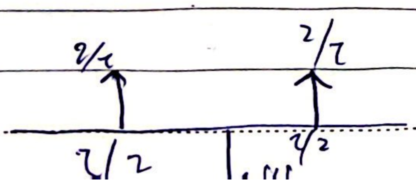
Example :-

Find the FT of triangular fn

$x(t) = \Delta\left(\frac{t}{\tau}\right)$



dx/dt



$\frac{d^2x}{dt^2}$

Date

SS - VAO

From the time shifting property.

$$\delta(t-t_0) \Leftrightarrow e^{-j\omega t_0}$$

$$\frac{d^2x}{dt^2} = \frac{2}{\tau}\left[\delta\left(t+\frac{\tau}{2}\right) - 2\delta(t) + \delta\left(t-\frac{\tau}{2}\right)\right].$$

Taking Fourier Transform.

$$-\omega^2 X(\omega) = \frac{2}{\tau}\left[e^{j\left(\frac{\omega\tau}{2}\right)} - 2 + e^{-j\left(\frac{\omega\tau}{2}\right)}\right].$$

$$= \frac{4}{\tau}\left(\cos\frac{\omega\tau}{2} - 1\right).$$

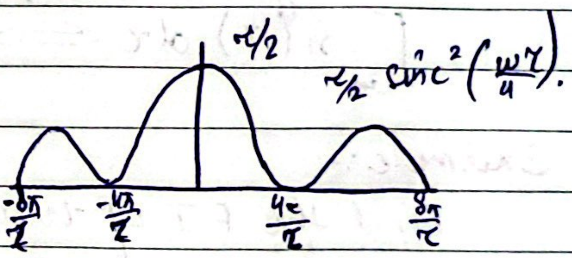
$$-\omega^2 X(\omega) = -\frac{8}{\tau}\sin^2\left(\frac{\omega\tau}{4}\right),$$

and

$$X(\omega) = \frac{8}{\omega^2\tau}\sin^2\left(\frac{\omega\tau}{4}\right).$$

$$= \frac{\tau}{2}\left[\frac{\sin\left(\omega\tau/4\right)}{\omega\tau/4}\right]^2$$

$$= \frac{\tau}{2}\,\mathrm{sinc}^2\left(\frac{\omega\tau}{4}\right).$$



This procedure can be applied to any function made up of straight line segment with

$$x(t) \to 0 \quad \text{as} \quad |t| \to \infty.$$

Fourier Transform Operations

$x(t) \Longleftrightarrow X$

$x(t-t_0) \Longleftrightarrow X(\omega$

		$x(t)$	$X(\omega)$
1.	Scalar Multiplication	$k\,x(t)$	$k\,X(\omega)$
2.	Addition	$x_1(t)+x_2(t)$	$X_1(\omega)+X_2(\omega)$
3.	Duality	$X(t)$	$2\pi\, x(-\omega)$
4.	Scaling	$x(at)$	$\frac{1}{\lvert a\rvert} X\left(\frac{\omega}{a}\right)$
5.	Time Shifting	$x(t-t_0)$	$X(\omega)\, e^{-j\omega t_0}$
6.	Frequency Shifting (ω_0 real)	$x(t)e^{j\omega_0 t}$	$X(\omega-\omega_0)$
7.	Time convolution	$x_1(t) * x_2(t)$	$X_1(\omega)X_2(\omega)$
8	Frequency convolution	$x_1(t)x_2(t)$	$\frac{1}{2\pi} X_1(\omega) * X_2(\omega)$

PCM = Pulse Code Modulation

To reconstruct $g(t)$ from $\bar{g}(t)$, $G(\omega)$ needs to be recovered from $\bar{G}(\omega)$. This is only possible when there is ~~only~~ no overlap b/w successive cycles of $\bar{G}(\omega)$, so

$$f_s > 2B$$

or $\left[T_s < \frac{1}{2B}\right]$

$\Rightarrow$ The min. sampling frequency $f_s = 2B$ required to recover $g(t)$ from its sample $\bar{G}(t)$ is called Nyquist rate for $g(t)$ and the corresponding sampling interval $T_s = \frac{1}{2B}$ is called Nyquist interval.

Q) A television signal, audio & video, has a bandwidth of 4.5 MHz. This signal is sampled, quantized and binary coded to obtain a PCM signal.

a) Determine sampling rate if signal is sampled at 20% above the Nyquist rate.

Sol) $B = 4.5\,\text{MHz}$

~~sampled, quantized and binary coded~~

100%. ← Nyquist rate $= 2B = 2 \times 4.5 = 9\,\text{MHz}$

20% above

NS = 120%. Sampling rate = 120% of 9.5 M~~Hz~~ = 10.8 M sample/second

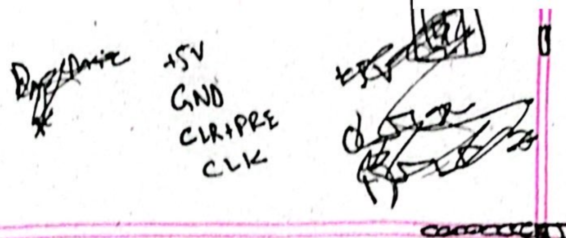
b) If sample are quantized into 1024 levels, determine the binary pulses required to encode each sample.

Sol) 1024 ~~total~~ levels $Q_0 - Q_{1023}$

$2^{10} = 1024$

10 binary pulses to encode each sample.

+5V
GND
CLR+PRE
CLK

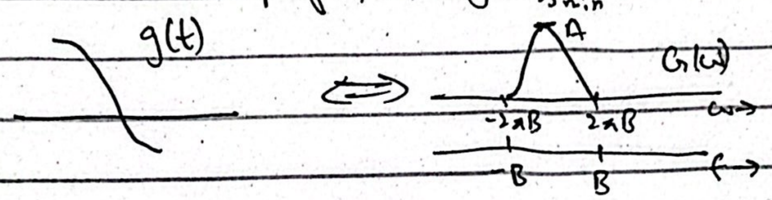


Sampling Theorem :-

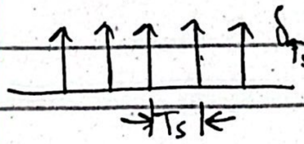
Any signal whose spectrum is bandlimited to B Hz $[G(\omega)=0$ for $|\omega|>2\pi B]$ can be reconstructed exactly (without errors) from its samples taken uniformly at a rate $R>2B$ samples/second.

Analog-to-Digital conversion of signals
1) Sampling (continuous to discrete) (x-axis)
2) Quantization. (y-axis)
3) Code Assignment (binary code)

Min. sampling frequency $= f_{s_{min}} = 2B\,Hz$

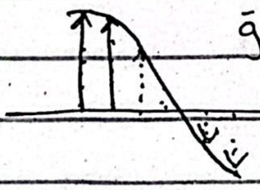


→ Multiplication with an impulse train



$$\delta_{T_s}(t)=\frac{1}{T_s}\left[1+2\cos\omega_s t+2\cos 2\omega_s t+2\cos 3\omega_s t+\dots\right]$$

$$\omega_s=\frac{2\pi}{T_s}=2\pi f_s$$



$$\bar{g}(t)=g(t)\,\delta_{T_s}(t)$$

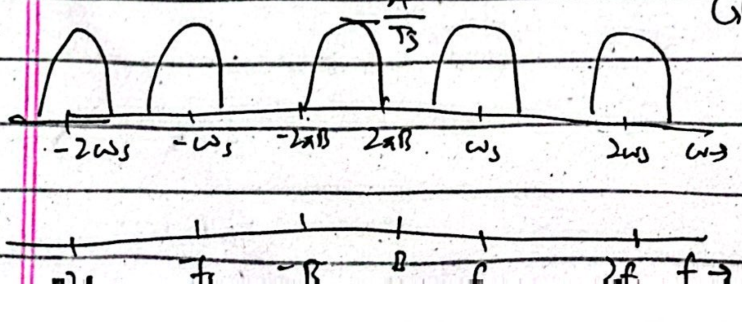
$$\bar{g}(t)=\frac{1}{T_s}\left[g(t)+2g(t)\cos\omega_s t+2g(t)\cos 2\omega_s t+2g(t)\cos 3\omega_s t+\dots\right]$$

$G(\omega)$

$G(\omega-\omega_s)+G(\omega+\omega_s)$

$G(\omega-2\omega_s)+G(\omega+2\omega_s)$

$$\bar{G}(\omega)=\frac{1}{T_s}\sum_{n=-\infty}^{\infty}G(\omega-n\omega_s)$$



Date

DAY-24

Time Shifting Property :-

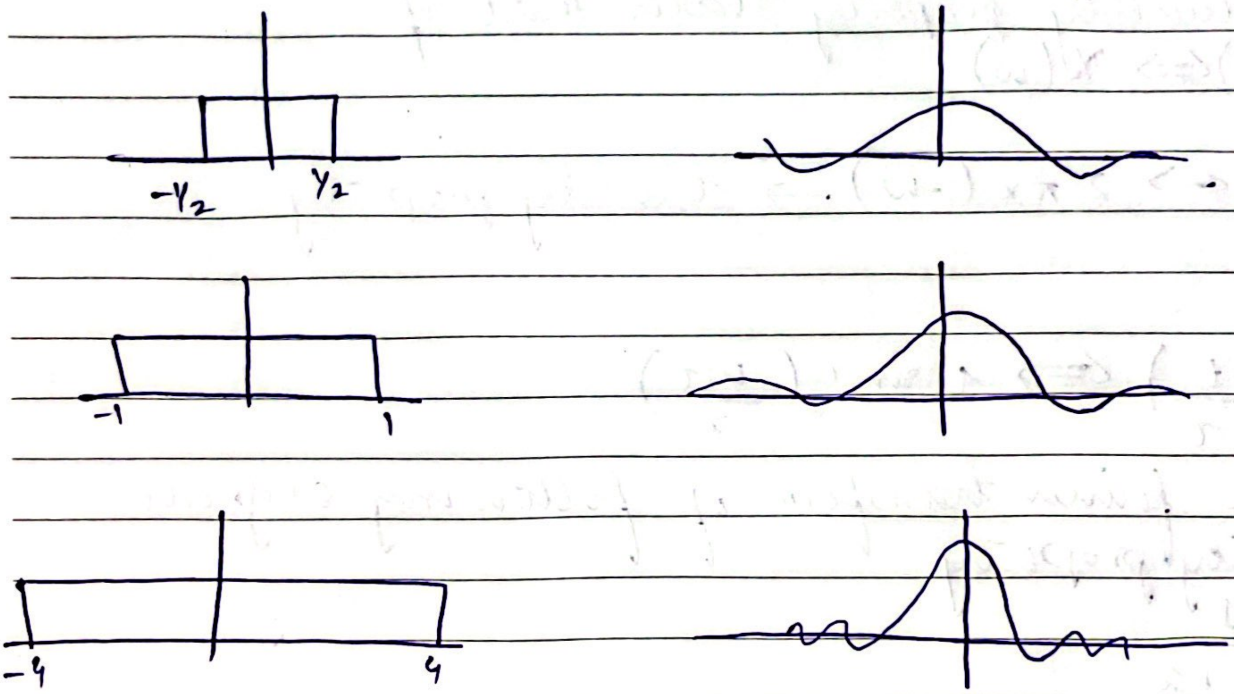
If

$$x(t) \Longleftrightarrow X(\omega)$$

then, for any real constant a,

$$x(at) \Longleftrightarrow \frac{1}{|a|} X\left(\frac{\omega}{a}\right)$$

- Time compression $(a>1)$ results in spectral expansion.
- Time expansion $(a<1)$ results in spectral compression.



Time reversal Property :-

Scaling the time variable 't' by

$$x(t) \Longleftrightarrow \cancel{X}(\omega)$$

then $x(-t) \Longleftrightarrow X(-\omega)$

Example :-

$$x(t) = e^{at}\, u(-t)$$

Date

DS-VAO

$x(t) = X(\omega)$

$x(t)$	$X(\omega)$	
$e^{-a}u(t$	$\frac{1}{a+j\omega}$	$a>0$
$e^{at}u(-t)$	$\frac{1}{a \mp j\omega}$	$a<0$

Duality Property :-

the duality property states that if
$x(t) \Leftrightarrow X(\omega)$
then
$X(t) \Leftrightarrow 2\pi x(-\omega) \rightarrow$ duality property

Example :-

$rect\left(\frac{t}{\tau}\right) \Leftrightarrow \tau \, sinc\left(\frac{\omega\tau}{2}\right)$

- Find the fourier transform of following signals using duality property

1) $x_1(t) = \frac{1}{jt+a}$

2) $x_2(t) = \pi\left[\delta(t-t_0) + \delta(t+t_0)\right]$

Solution:- (1)

$e^{-at}u(t) \longleftrightarrow \frac{1}{a+j\omega}$

Date

- $\frac{1}{jt+a} \longleftrightarrow 2\pi e^{aw} u(w)$

  $\cos w_0 t \longleftrightarrow \pi[\delta(w-w_0) + \delta(w+w_0)$

- $\pi[\delta(t-t_0) + \delta(t-t_0)] \longleftrightarrow 2\pi \cos t_0 w$

Differentiation Property

If

$$x(t) \longleftrightarrow X(w)$$

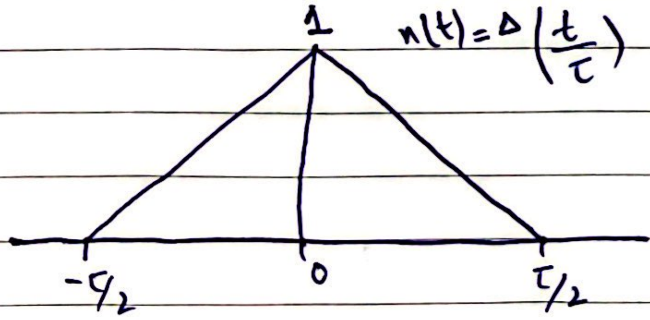
then

$$\frac{dx(t)}{dt} \longleftrightarrow jwX(w)$$

- Effect of differentiation in time domain is multiplication of $X(w)$ by jw in frequency domain
- Repeated application of this property ~~will~~ yields

$$\frac{d^n x}{dt^n} \longleftrightarrow (jw)^n X(w)$$

Example:-



Date

from the time-differentiation property

$$\frac{d^2x}{dt^2} \longleftrightarrow (j\omega)^2 X(\omega) = -\omega^2 X(\omega)$$

Also, from the time-shifting property

$$\delta(t-t_0) \longleftrightarrow e^{-j\omega_0}$$

$$\frac{d^2x}{dt^2} = \frac{2}{\tau}\left[\delta\left(t+\frac{\tau}{2}\right) - 2\,\delta(t) + \delta\left(t-\frac{\tau}{2}\right)\right]$$

$$-\omega^2 X(\omega) = \frac{2}{\tau}\left[e^{j(\omega t/2)} - 2 + e^{-j(\omega t/2)}\right]$$

$$= \frac{4}{\tau}\left(\cos\frac{\omega\tau}{2} - 1\right)$$

$$= \frac{-8}{\tau}\sin^2\left(\frac{\omega\tau}{4}\right)$$

$$X(\omega) = \frac{8}{\omega^2\tau}\sin^2\left(\frac{\omega\tau}{4}\right) = \frac{\tau}{2}\left[\frac{\sin\left(\frac{\omega\tau}{4}\right)}{\frac{\omega\tau}{4}}\right]^2 = \frac{\tau}{2}\,\mathrm{sinc}^2\left(\frac{\omega\tau}{4}\right)$$

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